Exercise 2

Donnerstag, 30. Oktober 2025 08:22

Exercise: Atmosphere and Airspeeds

The indicated airspeed of the aircraft is $V_{IAS} = 320{,}8\ kts$. The atmosphere is assumed to be the standard atmosphere. The density of the surrounding air is $\tilde{\rho} = 0{,}916\ \frac{kg}{m^3}$.

For this type of aircraft, the following relation holds: $V_{CAS} = 1{,}01 \cdot (V_{IAS} - 2\ kts)$.

Please use the following values in your calculation:

MSL density according to ISA standard atmosphere:	$\rho_{s,MSL} = 1.225\ \frac{kg}{m^3}$
MSL pressure according to ISA standard atmosphere:	$p_{s,MSL} = 1.01325e5\ Pa$
MSL temperature according to ISA standard atmosphere:	$T_{s,MSL} = 288.15\ K$
Gravitational acceleration:	$g_0 = 9.80665\ \frac{m}{s^2}$
MSL speed of sound:	$a_{MSL} = 340\ \frac{m}{s}$
Specific gas constant:	$R = 287.05\ \frac{J}{kg\,K}$
Polytropic exponent:	$n = 1.235$
Isentropic exponent:	$\kappa = 1{,}4$
Radius of the earth:	$r_E = 6356766m$

a) What is the geometric height of the aircraft?

$$\frac{\rho}{\rho_{MSL}} = \left(1 + \frac{\gamma_{TR}}{T_{s,MSL}} \cdot H_G\right)^{\frac{1}{\mu_{TR}-1}}$$

$$H_G = \frac{r_E \cdot h}{r_E + h} \rightarrow h = \frac{H_G \cdot r_E}{r_E - H_G} = 29285\ m;$$

$$\left(\frac{\rho}{\rho_{MSL}}\right)^{\mu_{TR}-1} = \left(1 + \frac{\gamma_{TR}}{T_{s,MSL}} \cdot H_G\right)$$

$$\rightarrow H_G = \frac{T_{s,MSL}}{\gamma_{TR}}\left(\left(\frac{\rho}{\rho_{MSL}}\right)^{\mu_{TR}-1} - 1\right) = 2927{,}1\ m;$$

b) Calculate the calibrated airspeed V_{CAS}.

$$V_{CAS} = 1.01\ (V_{TAS} - 2\ kts) = 321{,}99\ kts = 165{,}64\ \frac{m}{s};$$

c) Can we assume the air to be incompressible?

Condition: $Ma < 0.3 \rightarrow$ Assume incompressible flow

$Ma = \frac{V}{a}$ ← TRUE Airspeed ← Speed of Sound

$a = \sqrt{\kappa \cdot R \cdot T(H_G)}$

$T(H_G) = T_{S,MSL} + \gamma \cdot H_G = 288{,}15\,K + (-0.0065\,\frac{K}{m} \cdot 2927{,}1\,m)$

$= 269.12\,K$

$a = 328.87\,\frac{m}{s}$;

$Ma_{CAS} = \frac{165.64\,m/s}{328.87\,m/s} \approx 0.5$; → Compressible! as $V_{TAS} > V_{CAS}$, $Ma > Ma_{CAS}$

d) Calculate the difference between total and static pressure.

$V_{CAS} = 156.64\,m/s$

from Table: $V_{CAS} = a_{MSL} \cdot \sqrt{\frac{2}{\kappa-1}\left(\left(\frac{\Delta p}{p_{MSL}} + 1\right)^{\frac{\kappa-1}{\kappa}} - 1\right)} = 156.64\,m/s$;

$\Delta p = \left(\left(\frac{V_{CAS}}{a_{MSL}}\right)^2 \cdot \frac{\kappa-1}{2} + 1\right)^{\frac{\kappa}{\kappa-1}} - 1 \cdot p_{MSL} = 17856{,}69\,Pa$;

e) Calculate the equivalent airspeed V_{EAS}.

$V_{EAS} = V_{TAS} \cdot \sqrt{\frac{\rho}{\rho_{MSL}}}$

$V_{CAS} = a \cdot \sqrt{\frac{2}{\kappa-1} \cdot \frac{\rho}{\rho_{MSL}}\left(\left(\frac{\Delta p}{p} + 1\right)^{\frac{\kappa-1}{\kappa}} - 1\right)}$

$\rho = 0.916\,kg/m^3$ from Task d

from Task c

p = ?

from ISA Equations:

$p(H_G) = p_{MSL}\left(1 + \frac{\alpha_{TR}}{T_{S,MSL}} \cdot H_G\right)^{\frac{g_0}{R \cdot \alpha_{TR}} - 1} = 7.0764 \cdot 10^4\,Pa$;

$p_{MSL} = 101325\,Pa$

from a): $H_G = 2927.1\,m$

$\Rightarrow V_{EAS} = 167.87\ m/s$;

f) Calculate the true airspeed V_{TAS}.

$$V_{TAS} = V_{EAS} \sqrt{\frac{\rho_{ISA}}{\rho}} = 189.5\ \frac{m}{s}$$;

(V_{EAS}: e; ρ: given)